

ISE 5405: Optimization I

Lecture 6: Simplex I

From a Basis to a Better Corner

BT Sections 3.1–3.2

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What You Should Be Able to Do

1. Explain why simplex moves between basic feasible solutions.
2. Construct a basic direction by solving one basis system.
3. Interpret a reduced cost as an objective slope.
4. Apply the ratio test and identify an entering and leaving variable.
5. Recognize optimality, unboundedness, and a degenerate pivot.

Today's route

See the path, derive the direction, price the direction, then take one pivot.

The Standard-Form Setting

We study

$$\min\{c^\top x : Ax = b, x \geq 0\}, \quad A \in \mathbb{R}^{m \times n}, \quad \text{rank}(A) = m.$$

A feasible basis chooses m independent columns $B = A_{\mathcal{B}}$ and sets

$$x_{\mathcal{B}} = B^{-1}b \geq 0, \quad x_{\mathcal{N}} = 0.$$

Starting point

Primal simplex starts from a feasible basis and preserves feasibility while it searches for a lower objective value.

The Simplex Method: The Big Picture

1. Start at a basic feasible solution, geometrically a corner.
2. Look along incident basic directions for a cost-reducing direction.
3. Move as far as feasibility permits to a neighboring corner.
4. Stop when no nonbasic direction has a negative reduced cost.

Why corners are enough

If a linear program has an optimal solution, then it has an optimal basic feasible solution.

How to Read the GILP Sequence

GILP means *Geometric Interpretation of Linear Programs*, an interactive LP and simplex visualizer. Its 2D sequence uses

$$\max z = 5x_1 + 3x_2,$$

whereas our algebra uses minimization. Multiplying the objective by -1 connects the conventions:

larger z in the picture



negative reduced cost for minimizing $-z$

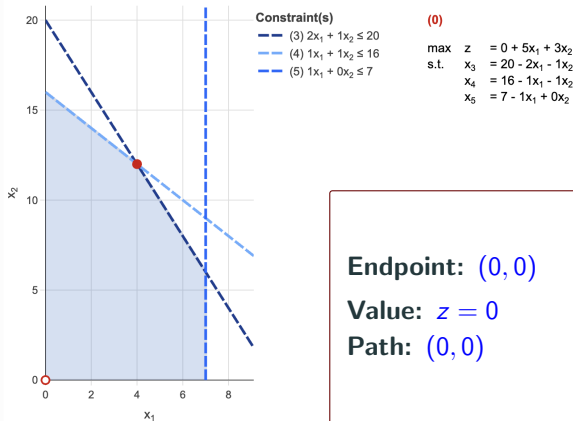
Watch three things

The red path accumulates, the current basis dictionary changes, and every move follows an edge of the blue feasible set.

GILP Path in Two Dimensions: Iteration 0 at (0,0)

Problem: $\max z = 5x_1 + 3x_2$

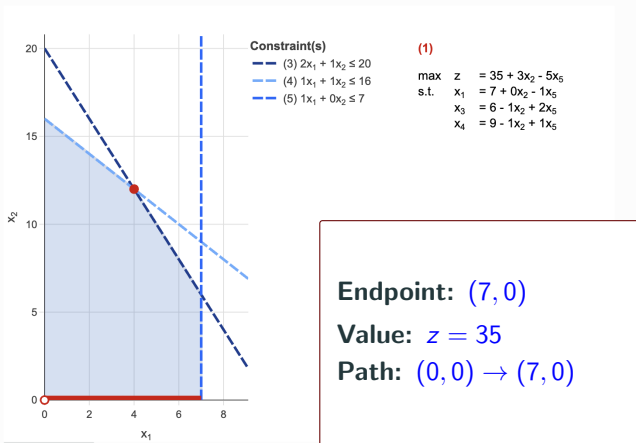
s.t. $2x_1 + x_2 \leq 20$, $x_1 + x_2 \leq 16$, $x_1 \leq 7$, $x_1, x_2 \geq 0$.



GILP Path in Two Dimensions: Iteration 1 at (7,0)

All Integer 2D LP

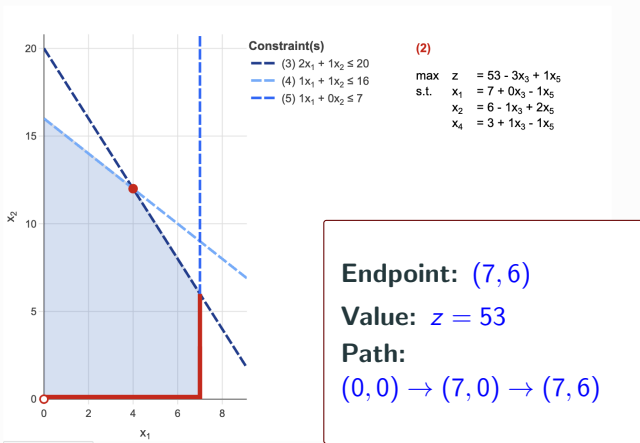
A 2D LP where all basic feasible solutions are integral and have integral tableaus.



GILP Path in Two Dimensions: Iteration 2 at (7,6)

All Integer 2D LP

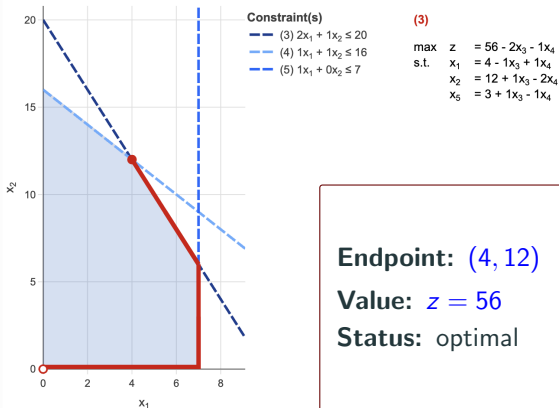
A 2D LP where all basic feasible solutions are integral and have integral tableaus.



GILP Path in Two Dimensions: Iteration 3 at (4, 12)

All Integer 2D LP

A 2D LP where all basic feasible solutions are integral and have integral tableaus.



What the 2D Path Shows

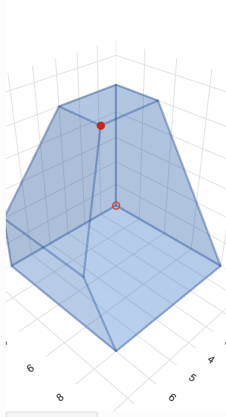
$$(0, 0) \longrightarrow (7, 0) \longrightarrow (7, 6) \longrightarrow (4, 12)$$

$$z : 0 \longrightarrow 35 \longrightarrow 53 \longrightarrow 56.$$

- Each red segment stays feasible and connects neighboring corners.
- Each dictionary expresses the objective and basic variables in terms of the current nonbasic variables.
- At the last corner, no available edge can increase z .

3D GILP Path: Iteration 0

Problem: $\max z = x_1 + 2x_2 + 4x_3$
s.t. $x_1 \leq 6, x_1 + x_3 \leq 8, x_3 \leq 5,$
 $x_2 + x_3 \leq 8, x_1, x_2, x_3 \geq 0.$



Constraint(s)

- (4) $1x_1 + 0x_2 + 0x_3 \leq 6$
- (5) $1x_1 + 0x_2 + 1x_3 \leq 8$
- (6) $0x_1 + 0x_2 + 1x_3 \leq 5$
- (7) $0x_1 + 1x_2 + 1x_3 \leq 8$

(0)

max $z = 0 + 1x_1 + 2x_2 + 4x_3$
s.t. $x_4 = 6 - 1x_1 + 0x_2 + 0x_3$
 $x_5 = 8 - 1x_1 + 0x_2 - 1x_3$
 $x_6 = 5 + 0x_1 + 0x_2 - 1x_3$
 $x_7 = 8 + 0x_1 - 1x_2 - 1x_3$

Endpoint: $(0, 0, 0)$

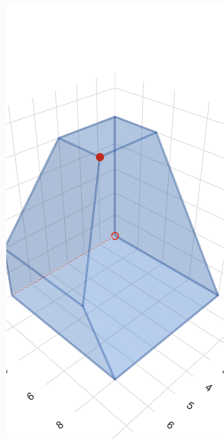
Value: $z = 0$

Path: origin

3D GILP Path: Iteration 1

All Integer 3D LP

A 3D LP where all basic feasible solutions are integral and have integral tableaus.



Constraint(s)

- (4) $1x_1 + 0x_2 + 0x_3 \leq 6$
- (5) $1x_1 + 0x_2 + 1x_3 \leq 8$
- (6) $0x_1 + 0x_2 + 1x_3 \leq 5$
- (7) $0x_1 + 1x_2 + 1x_3 \leq 8$

(1)

$$\begin{array}{ll} \max & z = 6 + 2x_2 + 4x_3 - 1x_4 \\ \text{s.t.} & x_1 = 6 + 0x_2 + 0x_3 - 1x_4 \\ & x_5 = 2 + 0x_2 - 1x_3 + 1x_4 \\ & x_6 = 5 + 0x_2 - 1x_3 + 0x_4 \\ & x_7 = 8 - 1x_2 - 1x_3 + 0x_4 \end{array}$$

Endpoint: (6, 0, 0)

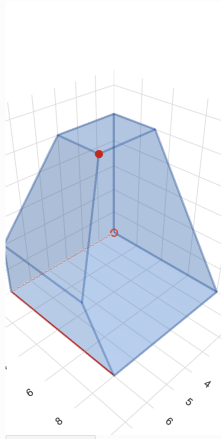
Value: $z = 6$

Step: first edge

3D GILP Path: Iteration 2

All Integer 3D LP

A 3D LP where all basic feasible solutions are integral and have integral tableaus.



Constraint(s)

- (4) $1x_1 + 0x_2 + 0x_3 \leq 6$
- (5) $1x_1 + 0x_2 + 1x_3 \leq 8$
- (6) $0x_1 + 0x_2 + 1x_3 \leq 5$
- (7) $0x_1 + 1x_2 + 1x_3 \leq 8$

(2)

$$\begin{array}{ll}\max & z = 22 + 2x_3 - 1x_4 - 2x_7 \\ \text{s.t.} & x_1 = 6 + 0x_3 - 1x_4 + 0x_7 \\ & x_2 = 8 - 1x_3 + 0x_4 - 1x_7 \\ & x_5 = 2 - 1x_3 + 1x_4 + 0x_7 \\ & x_6 = 5 - 1x_3 + 0x_4 + 0x_7\end{array}$$

Endpoint: (6, 8, 0)

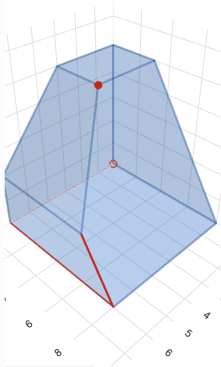
Value: $z = 22$

Step: second edge

3D GILP Path: Iteration 3

All Integer 3D LP

A 3D LP where all basic feasible solutions are integral and have integral tableaus.



Constraint(s)

$$(4) \ 1x_1 + 0x_2 + 0x_3 \leq 6$$

$$(5) \ 1x_1 + 0x_2 + 1x_3 \leq 8$$

$$(6) \ 0x_1 + 0x_2 + 1x_3 \leq 5$$

$$(7) \ 0x_1 + 1x_2 + 1x_3 \leq 8$$

(3)

$$\begin{array}{ll} \max & z = 26 + 1x_4 - 2x_5 - 2x_7 \\ \text{s.t.} & x_1 = 6 - 1x_4 + 0x_5 + 0x_7 \\ & x_2 = 6 - 1x_4 + 1x_5 - 1x_7 \\ & x_3 = 2 + 1x_4 - 1x_5 + 0x_7 \\ & x_6 = 3 - 1x_4 + 1x_5 + 0x_7 \end{array}$$

Endpoint: (6, 6, 2)

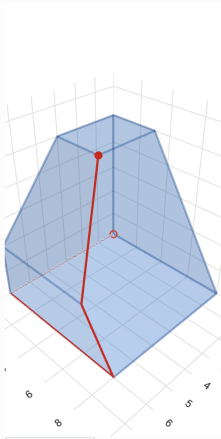
Value: $z = 26$

Step: third edge

3D GILP Path: Iteration 4

All Integer 3D LP

A 3D LP where all basic feasible solutions are integral and have integral tableaus.



Constraint(s)

$$(4) \ 1x_1 + 0x_2 + 0x_3 \leq 6$$

$$(5) \ 1x_1 + 0x_2 + 1x_3 \leq 8$$

$$(6) \ 0x_1 + 0x_2 + 1x_3 \leq 5$$

$$(7) \ 0x_1 + 1x_2 + 1x_3 \leq 8$$

(4)

$$\begin{array}{ll} \max & z = 29 - 1x_5 - 1x_6 - 2x_7 \\ \text{s.t.} & x_1 = 3 - 1x_5 + 1x_6 + 0x_7 \\ & x_2 = 3 + 0x_5 + 1x_6 - 1x_7 \\ & x_3 = 5 + 0x_5 - 1x_6 + 0x_7 \\ & x_4 = 3 + 1x_5 - 1x_6 + 0x_7 \end{array}$$

Endpoint: $(3, 3, 5)$

Value: $z = 29$

Status: final state

The Same Mechanism in 3D

- The blue polyhedron replaces the 2D polygon.
- The cumulative red path still follows one edge at a time.
- A pivot changes the basis and dictionary, not the feasible set.
- The algebra does not depend on our ability to draw the geometry.

The algorithmic question

At the current basis, which nonbasic variable gives an improving direction, and how far can we move along it?

Feasible Directions

Definition

For $x \in P$, a vector $d \in \mathbb{R}^n$ is a **feasible direction** at x if some $\epsilon > 0$ satisfies

$$x + \theta d \in P \quad \text{for every } 0 \leq \theta \leq \epsilon.$$

For $P = \{x : Ax = b, x \geq 0\}$, two checks are required:

$$Ad = 0 \quad \text{and} \quad x + \theta d \geq 0 \text{ for a small positive step.}$$

A Basis Partitions the Variables

For a feasible basis $B = A_{\mathcal{B}}$, write

$$A = [B \ A_{\mathcal{N}}], \quad x = \begin{bmatrix} x_{\mathcal{B}} \\ x_{\mathcal{N}} \end{bmatrix}, \quad x_{\mathcal{B}} = B^{-1}b, \quad x_{\mathcal{N}} = 0.$$

To leave this corner, choose one nonbasic variable x_j to increase while the other nonbasic variables remain zero.

One column, one candidate edge

Each nonbasic column A_j defines one basic direction relative to the current basis.

Construct a Basic Direction: State 1 of 3

Choose a nonbasic index $j \in \mathcal{N}$. Prescribe

$$d_j = 1, \quad d_k = 0 \quad (k \in \mathcal{N} \setminus \{j\}).$$

The unknown components are the basic components $d_{\mathcal{B}}$.

Interpretation

x_j increases at unit rate; the basic variables must compensate so that the equality constraints remain satisfied.

Construct a Basic Direction: State 2 of 3

Keep the prescribed nonbasic components visible:

$$d_j = 1, \quad d_k = 0 \quad (k \in \mathcal{N} \setminus \{j\}).$$

Equality preservation requires

$$0 = Ad = Bd_{\mathcal{B}} + A_j.$$

Thus the basic components solve the basis system

$$Bd_{\mathcal{B}} = -A_j.$$

Construct a Basic Direction: State 3 of 3

The complete cumulative derivation is

$$d_j = 1, \quad d_k = 0 \ (k \in \mathcal{N} \setminus \{j\}),$$

$$0 = Ad = Bd_{\mathcal{B}} + A_j, \quad \boxed{d_{\mathcal{B}} = -B^{-1}A_j}.$$

Compute by solving

Solve $Bd_{\mathcal{B}} = -A_j$. The inverse notation describes the answer; it is not an instruction to form B^{-1} .

Equality Preservation Is Not Enough

Nondegenerate basic feasible solution

If $x_B > 0$, every basic direction permits a sufficiently small positive step: $x_B + \theta d_B \geq 0$.

Degenerate basic feasible solution

If some $x_{B(i)} = 0$ and $d_{B(i)} < 0$, then every positive step makes that component negative. The direction is blocked at $\theta = 0$.

Degeneracy separates an algebraically valid direction from a direction that permits a positive geometric move.

Objective Slope Along a Basic Direction

Along $x(\theta) = x + \theta d$,

$$c^\top x(\theta) = c^\top x + \theta c^\top d.$$

For the j -th basic direction,

$$\begin{aligned} c^\top d &= c_j + c_B^\top d_B \\ &= c_j - c_B^\top B^{-1} A_j. \end{aligned}$$

This scalar is the rate of objective change per unit increase in x_j .

Definition

The reduced cost of variable x_j relative to basis B is

$$\bar{c}_j = c_j - c_B^\top B^{-1} A_j.$$

For a minimization problem:

- $\bar{c}_j < 0$: the j -th basic direction has an improving slope;
- $\bar{c}_j = 0$: the objective is locally flat in that direction;
- $\bar{c}_j > 0$: that direction initially raises the objective.

Reduced Costs Are the Objective-Row Coefficients

Use the current basis B and nonbasic columns A_N :

$$f = c_B^\top x_B + c_N^\top x_N.$$

The constraints give the dictionary

$$Bx_B + A_Nx_N = b \implies x_B = B^{-1}b - B^{-1}A_Nx_N.$$

Substitute into the objective and collect terms:

$$\begin{aligned} f &= c_B^\top (B^{-1}b - B^{-1}A_Nx_N) + c_N^\top x_N \\ &= c_B^\top B^{-1}b + \sum_{j \in N} \underbrace{(c_j - c_B^\top B^{-1}A_j)}_{\bar{c}_j} x_j. \end{aligned}$$

Same example after x_1 enters

$x_1 = 7 - x_5$ gives $f = -5(7 - x_5) - 3x_2 = -35 - 3x_2 + 5x_5$. Thus $\bar{c}_2 = -3$ and $\bar{c}_5 = 5$, relative to the new basis.

Every Basic Variable Has Zero Reduced Cost

For a basic index $B(i)$,

$$B^{-1}A_{B(i)} = e_j.$$

Therefore

$$\begin{aligned}\bar{c}_{B(i)} &= c_{B(i)} - c_B^\top B^{-1}A_{B(i)} \\ &= c_{B(i)} - c_B^\top e_j \\ &= 0.\end{aligned}$$

Only nonbasic reduced costs need to be examined.

Theorem

Let x be a basic feasible solution associated with basis B .

1. If every reduced cost is nonnegative, then x is optimal.
2. If x is optimal and nondegenerate, then every reduced cost is nonnegative.

Degenerate caveat

An optimal degenerate point can have a basis with a negative reduced cost whose direction is blocked at step zero.

Why Nonnegative Reduced Costs Prove Optimality

For any feasible x ,

$$x_B = B^{-1}b - B^{-1}A_N x_N.$$

Substitute into the objective:

$$\begin{aligned} c^\top x &= c_B^\top B^{-1}b + (c_N^\top - c_B^\top B^{-1}A_N) x_N \\ &= c_B^\top B^{-1}b + \bar{c}_N^\top x_N. \end{aligned}$$

If $\bar{c}_N \geq 0$ and $x_N \geq 0$, no feasible point has value below the current basic value.

The Two Checks for an Optimal Basis

A basis is optimal when it passes both tests:

$$\boxed{B^{-1}b \geq 0} \quad \text{and} \quad \boxed{\bar{c}_N \geq 0}.$$

Primal feasibility: the associated basic solution lies in the feasible set.

Optimality: no nonbasic direction has a negative objective slope.

Keep the questions separate

A nonsingular basis need not be feasible, and a feasible basis need not be optimal.

A Negative Reduced Cost Chooses an Entering Variable

Suppose a nonbasic x_j has $\bar{c}_j < 0$. Let

$$u = B^{-1}A_j, \quad d_B = -u, \quad d_j = 1.$$

Along the candidate path,

$$x_B(\theta) = x_B - \theta u, \quad c^\top x(\theta) = c^\top x + \theta \bar{c}_j.$$

Variable x_j **enters** the basis if the ratio test permits the move.

The Maximum Feasible Step

For each component with $u_i > 0$, nonnegativity requires

$$x_{\mathcal{B},i} - \theta u_i \geq 0 \implies \theta \leq \frac{x_{\mathcal{B},i}}{u_i}.$$

Thus

$$\theta^* = \min_{i: u_i > 0} \frac{x_{\mathcal{B},i}}{u_i}.$$

An index attaining the minimum identifies a basic variable that reaches zero and may **leave** the basis.

When No Variable Blocks the Move

If $u \leq 0$, then $d_B = -u \geq 0$, so

$$x + \theta d \geq 0 \quad \text{for every } \theta \geq 0.$$

Moreover,

$$Ad = 0, \quad d \geq 0, \quad c^\top d = \bar{c}_j < 0.$$

Unboundedness certificate

$x + \theta d$ stays feasible while its objective tends to $-\infty$. The vector d certifies that the minimization problem is unbounded.

One Simplex Iteration: State 1 of 2

1. Start with a feasible basis B and $x_B = B^{-1}b$.
2. Compute the coefficients in the objective row for the nonbasic variables:

$$\bar{c}_j = c_j - c_B^\top B^{-1}A_j.$$

3. If every $\bar{c}_j \geq 0$, stop: the current basis is optimal.
4. Otherwise choose an entering j with $\bar{c}_j < 0$ and solve $Bu = A_j$.

One Simplex Iteration: State 2 of 2

1. Start with a feasible basis B and $x_B = B^{-1}b$.
2. Price the nonbasic columns; stop if every $\bar{c}_j \geq 0$.
3. Choose an entering j with $\bar{c}_j < 0$; solve $Bu = A_j$.
4. If $u \leq 0$, return the unbounded direction d .
5. Otherwise compute $\theta^* = \min_{i: u_i > 0} x_{B,i} / u_i$.
6. Choose a minimizing row, exchange the entering and leaving columns, and update the basic solution.

Worked Example: The Initial Dictionary

Minimize f subject to the three slack equations below and $x \geq 0$:

$$x_4 = 20 - x_1 - 2x_2 - 2x_3,$$

$$x_5 = 20 - 2x_1 - x_2 - 2x_3,$$

$$x_6 = 20 - 2x_1 - 2x_2 - x_3,$$

$$f = -10x_1 - 12x_2 - 12x_3.$$

With $B_0 = [A_4 \ A_5 \ A_6] = I$, set $x_1 = x_2 = x_3 = 0$:

$$x^{(0)} = (0, 0, 0, 20, 20, 20), \quad f^{(0)} = 0.$$

Rule for all three pivots. Choose the smallest-index nonbasic variable with negative reduced cost. Among minimum-ratio ties, choose the smallest-index basic variable to leave.

Pivot 1: Choose the Direction and Step

Initially $\bar{c}_1 = -10$, $\bar{c}_2 = \bar{c}_3 = -12$. Our rule selects x_1 to enter.

$$d_B = -B_0^{-1}A_1 = (-1, -2, -2)^T, \quad d_1 = 1.$$

Your turn: increase $x_1 = \theta$, keeping $x_2 = x_3 = 0$. Which variable leaves?

Basic variable	Along the move	Step bound
x_4	$20 - \theta$	$\theta \leq 20$
x_5	$20 - 2\theta$	$\theta \leq 10$
x_6	$20 - 2\theta$	$\theta \leq 10$

$\theta^* = \min\{20, 10, 10\} = 10$. The tie rule makes x_5 leave.

After the move. $x^{(1)} = (10, 0, 0, 10, 0, 0)$, with $f^{(1)} = -100$.

Both x_5 and x_6 reach zero, but only x_5 leaves the basis.

After Pivot 1: A Degenerate Basic Solution

Solve the leaving row for x_1 , then substitute into every other row, including f :

$$x_5 = 20 - 2x_1 - x_2 - 2x_3 \implies x_1 = 10 - \frac{1}{2}x_2 - x_3 - \frac{1}{2}x_5.$$

New basis $\mathcal{B} = (x_1, x_4, x_6)$; nonbasic variables $\mathcal{N} = (x_2, x_3, x_5)$.

$$x_1 = 10 - \frac{1}{2}x_2 - x_3 - \frac{1}{2}x_5,$$

$$x_4 = 10 - \frac{3}{2}x_2 - x_3 + \frac{1}{2}x_5,$$

$$x_6 = -x_2 + x_3 + x_5,$$

$$f = -100 - 7x_2 - 2x_3 + 5x_5.$$

Degenerate arrival after a positive step. Setting the nonbasic variables to zero gives $x_6 = 0$, although x_6 is basic. Next, our rule selects x_2 , since $\bar{c}_2 = -7 < 0$.

Pivot 2: Change the Basis with a Zero Step

Let $x_2 = \theta$, keeping $x_3 = x_5 = 0$. The current dictionary gives

$$x_1 = 10 - \frac{1}{2}\theta, \quad x_4 = 10 - \frac{3}{2}\theta, \quad x_6 = -\theta.$$

$\theta^* = \min\{20, 20/3, 0\} = 0$: x_2 **enters** and x_6 **leaves**.

Rearrange $x_6 = -x_2 + x_3 + x_5$ as $x_2 = x_3 + x_5 - x_6$, then substitute:

$$x_1 = 10 - \frac{3}{2}x_3 - x_5 + \frac{1}{2}x_6,$$

$$x_2 = x_3 + x_5 - x_6,$$

$$x_4 = 10 - \frac{5}{2}x_3 - x_5 + \frac{3}{2}x_6,$$

$$f = -100 - 9x_3 - 2x_5 + 7x_6.$$

New basis, same point: $\mathcal{B} = (x_1, x_2, x_4)$, $x^{(2)} = x^{(1)}$,

$f^{(2)} = -100$. Now x_2 is basic at zero. The objective slope is negative, but the feasible step is zero.

Pivot 3: Reach the Final Dictionary

Choose x_3 ($\bar{c}_3 = -9 < 0$). Set $x_3 = \theta$, $x_5 = x_6 = 0$.

$$x_1 = 10 - \frac{3}{2}\theta, \quad x_2 = \theta, \quad x_4 = 10 - \frac{5}{2}\theta.$$

x_2 increases, so only x_1, x_4 bound the step:

$\theta^* = \min\{20/3, 4\} = 4$. x_4 **leaves**.

Solve for $x_3 = 4 - \frac{2}{5}x_4 - \frac{2}{5}x_5 + \frac{3}{5}x_6$ and substitute:

$$x_1 = 4 + \frac{3}{5}x_4 - \frac{2}{5}x_5 - \frac{2}{5}x_6,$$

$$x_2 = 4 - \frac{2}{5}x_4 + \frac{3}{5}x_5 - \frac{2}{5}x_6,$$

$$x_3 = 4 - \frac{2}{5}x_4 - \frac{2}{5}x_5 + \frac{3}{5}x_6,$$

$$f = -136 + \frac{18}{5}x_4 + \frac{8}{5}x_5 + \frac{8}{5}x_6.$$

Set $x_4 = x_5 = x_6 = 0$: $x^{(3)} = (4, 4, 4, 0, 0, 0)$, with basis $\mathcal{B} = (x_1, x_2, x_3)$.

Verify Optimality and Review the Three Pivots

Feasible: $x^* = (4, 4, 4, 0, 0, 0)$ satisfies $Ax^* = b$, $x^* \geq 0$.

$$f = -136 + \underbrace{\frac{18}{5}}_{\bar{c}_4} x_4 + \underbrace{\frac{8}{5}}_{\bar{c}_5} x_5 + \underbrace{\frac{8}{5}}_{\bar{c}_6} x_6 \geq -136.$$

Since $x_4, x_5, x_6 \geq 0$, no feasible point beats -136 . Our point attains the bound: **optimal**.

Pivot	Enters / leaves	θ^*	New basis	f
1	x_1 / x_5	10	(x_1, x_4, x_6)	-100
2	x_2 / x_6	0	(x_1, x_2, x_4)	-100
3	x_3 / x_4	4	(x_1, x_2, x_3)	-136

Three pivots, two positive moves: a zero step changes the basis but preserves the point and objective.

Why Simplex Terminates Without Degeneracy

Assume every basic feasible solution is nondegenerate.

- Every finite pivot has $\theta^* > 0$ and $\bar{c}_j < 0$, so the objective strictly decreases.
- A basic feasible solution cannot be visited twice.
- There are only finitely many bases.

Hence simplex terminates after finitely many pivots with either an optimal basis or an unboundedness certificate.

What the assumption buys

Strict improvement, not merely a change of basis, drives the finite argument.

Degeneracy and Cycling: A Preview

At a degenerate basic feasible solution:

- the minimum ratio can be zero;
- the basis can change while the point and objective stay fixed;
- a sequence of such pivots can revisit a basis and cycle.

Pivot-rule precision

Choosing the smallest index only for a leaving tie is not enough. Bland's anti-cycling rule selects the smallest eligible index for both entering and leaving choices.

We will study degeneracy and anti-cycling rules in the later simplex meetings.

Implementation Preview and Takeaway

Full tableau

Stores $B^{-1}b$ and $B^{-1}A$; pivots are explicit row operations.

Revised simplex

Stores a basis factorization, computes only needed solves, and exploits sparsity. It does not repeatedly form B^{-1} .

The complete logic

feasible basis $\rightarrow$ price $\rightarrow$ direction
 $\rightarrow$ ratio test $\rightarrow$ pivot or certificate